Vaccine Pharmacology & Patient-Education Reference

Four practice questions for Alberta pharmacists, 2026–27 respiratory season, plus patient counselling sheets on duration of protection and on this year's strains. Prepared 29 October 2026; updated 2 November 2026. Evidence is drawn preferentially from peer-reviewed journals (NEJM, Lancet, JAMA Network Open, Clinical Infectious Diseases, Journal of Clinical Oncology, Science, Nature, CCDR); WHO technical reports and Alberta Immunization Policy documents are used only where they are the primary source (strain composition, provincial eligibility). Bracketed numbers refer to the reference list.

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1. Ipsilateral deltoid injection after axillary lymph-node removal

Bottom line

It is a **precaution, not a contraindication**. Three separate rationales are usually bundled together: (a) lymphedema risk in a limb with reduced lymphatic capacity, (b) vaccine-induced reactive axillary lymphadenopathy confounding breast-cancer surveillance, and (c) a theoretical immunologic concern. Only (b) is well supported; prospective data do not show that skin punctures or injections in the at-risk arm increase lymphedema, and there is no evidence that vaccines are less effective after node dissection. Use the contralateral deltoid when one is available; if nodes have been removed on both sides, use the vastus lateralis. Never withhold or delay a vaccine over arm choice.

1.1 Anatomy and physiology behind the precaution

An intramuscular deltoid injection drains through lymphatics of the lateral arm to the axillary node basin (levels I–III) and, in part, to the supraclavicular nodes. Axillary lymph-node dissection (ALND) removes most of that basin; regional nodal radiotherapy (RLNR) fibroses what remains. The limb is then functioning on reduced lymphatic transport capacity, and breast-cancer-related lymphedema (BCRL) develops when lymph load exceeds that capacity. In the largest prospective screening cohort, ALND, RLNR, BMI ≥25 kg/m² and cellulitis in the at-risk arm were the factors that significantly raised arm volume [1]. Lymphedema prevalence after ALND is roughly 20% and rises with the number of nodes removed and with radiotherapy [2].

The vaccine-specific concern is that a reactogenic injection creates a local inflammatory load (oedema, cytokine-driven hyperaemia, reactive lymphadenitis) draining into a compromised basin, and that the needle track is a potential portal for cellulitis, which is a proven trigger [1,3]. Sentinel-node biopsy alone (typically 1–3 nodes) carries a much lower baseline BCRL risk than ALND, so the strength of the precaution scales with the extent of axillary surgery and radiotherapy [2,3].

1.2 What the lymphedema evidence actually shows

- **Ferguson et al., J Clin Oncol 2016** [1]: 632 women undergoing breast-cancer treatment were screened prospectively with optoelectronic perimetry (3,041 bilateral arm-volume measurements). Ipsilateral blood draws (P = 0.62), injections (P = 0.77), blood-pressure readings (P = 0.39) and air travel (P = 0.77) were not associated with increases in relative arm volume; cellulitis, ALND, RLNR and BMI ≥25 were.
- **Kilbreath et al., Breast 2016** [4]: a large Australian prospective cohort likewise found no association between skin punctures in the at-risk arm and lymphoedema; axillary surgery extent, radiotherapy and body-mass were the drivers.
- **Asdourian et al., Lancet Oncol 2016** [3]: a systematic appraisal of the traditional precautions (air travel, ipsilateral blood pressure, skin puncture, extreme temperatures) concluded that the restrictions rest on expert opinion and historical case series rather than controlled data, while cellulitis prevention is the one intervention clearly supported.
- **Hadjistyllis et al., Ann Transl Med 2023** [5]: systematic review of skin-puncturing procedures after breast or axillary surgery; concluded that venous-access procedures can be performed safely in the ipsilateral arm.
- A 2022 San Antonio Breast Cancer Symposium analysis of 2,018 prospectively screened patients found no increase in BCRL with ipsilateral intravenous infusions (HR 0.85, P = 0.60) [6].

Evidence quality: moderate for "no association" (large prospective cohorts with objective volumetry; exposure self-reported, so recall bias and residual confounding remain). No randomized trial exists and none is likely. Isolated case reports of oedema after vaccination in the at-risk limb exist [7], which is why most immunization manuals still default to the contralateral side.

1.3 The rationale that is well supported: reactive lymphadenopathy and cancer surveillance

Intramuscular vaccines routinely produce reactive lymphadenopathy in the draining axilla. In the mRNA-1273 phase 3 trial, palpable axillary swelling or tenderness was reported by 11.6% after dose 1 and 16.0% after dose 2 [8]; on ^{18}F-FDG PET/CT, avid ipsilateral axillary nodes were seen in 45–54% of oncology patients imaged in the weeks after vaccination [9,10], and the same phenomenon is described after influenza and HPV vaccines [11]. In a patient with prior breast cancer, a new ipsilateral axillary node is indistinguishable from recurrence without additional imaging or biopsy, and in a newly diagnosed patient it can confound sentinel-node mapping.

For that reason the European Society of Breast Imaging (EUSOBI) recommends that patients with a history of breast cancer receive vaccines in the contralateral arm or anterolateral thigh [8,12], and the Canadian Society of Breast Imaging / Canadian Association of Radiologists advise that women with prior lumpectomy, node dissection, post-treatment lymphedema, melanoma or any axilla-draining cancer consider vaccination on the unaffected side or thigh "to avoid false positives and unknown implications on SLNB" [13]. Reactive nodes usually normalize within about 12 weeks; both societies advise against delaying either screening or vaccination [12,13].

1.4 The immunologic question: will the vaccine still work?

Vaccine-specific germinal-centre responses form in the lymph nodes draining the injection site; fine-needle aspirates of ipsilateral axillary nodes after mRNA vaccination show spike-binding germinal-centre B cells persisting for months [14]. After ALND the deltoid still drains, but through rerouted collateral channels to residual axillary, supraclavicular or contralateral nodes. There are **no clinical data** showing reduced seroconversion or effectiveness when a vaccine is given in an arm with prior ALND, and systemic antibody responses do not depend on any single nodal basin. The honest position for patient education is that the arm choice protects the limb and the imaging, not the vaccine's efficacy; if the at-risk arm is the only option, the vaccine should still be given.

1.5 Practice points

- Unilateral ALND or SLNB → contralateral deltoid (default).
- Bilateral axillary surgery, or arm unusable → vastus lateralis (anterolateral thigh), an accepted adult IM site in the Canadian Immunization Guide [15]; use a 25 mm (1-inch) needle or longer as for the deltoid.
- Established lymphedema → avoid the affected limb; any break in skin integrity raises cellulitis risk [1,3].
- Document the site; if the patient is awaiting breast imaging or sentinel-node mapping, note the date so radiology can interpret nodes correctly [13].
- Counsel on cellulitis warning signs (spreading redness, warmth, fever) rather than on injections per se.

Patient-education script

"Because some of the lymph nodes under your arm were removed, that arm has less drainage. A vaccine causes a small amount of swelling and often makes nearby lymph nodes react, so we use the other arm to be safe and to keep your follow-up scans clear. Studies that tracked hundreds of women found that needles in the treated arm did not actually cause lymphedema, so if we ever had to use that arm the vaccine would still work and the risk is low. The one thing that does raise lymphedema risk is a skin infection, so let us know about any redness or warmth after the shot."

2. The 2026–27 Alberta influenza vaccine: strains and how they were chosen

2.1 Composition

All three components were updated from 2025–26. The WHO recommendation of 27 February 2026 [16,17] distinguishes egg-propagated from cell-culture/recombinant prototypes; Alberta's Fluzone, Fluviral, Fluad and Fluzone High-Dose are egg-based, while Flucelvax is cell-based [18].

Component	Egg-based vaccines (Fluzone, Fluviral, Fluad, Fluzone HD)	Cell-/recombinant-based (Flucelvax)	Clade / why it changed
A(H1N1)pdm09	A/Missouri/11/2025-like	A/Missouri/11/2025-like	Clade 5a.2a.1, subclade D.3.1; D.3.1/D.3.1.1 viruses now predominate and human post-vaccination titres to them were significantly reduced versus the 2025–26 A/Wisconsin/67/2022 reference [17].
A(H3N2)	A/Darwin/1454/2025-like	A/Darwin/1415/2025-like	Subclade K (formerly J.2.4.1; HA K2N, S144N, N158D, I160K, Q173R, T328A, S378N). Emerged August 2025, drove an early, intense season; ferret antisera to the 2025–26 A/Croatia/10136RV/2023 virus recognized K viruses poorly (HI titres <40 vs 640–1280 with Darwin antisera) [16,17].
B/Victoria	B/Tokyo/EIS13-175/2025-like	B/Pennsylvania/14/2025-like	Clade 3a.2, subclade C.3.1 with HA D197N (new N-glycosylation site); C.3/C.3.1 viruses rising in North America and East Asia and poorly inhibited by B/Austria/1359417/2021 antisera [17].
B/Yamagata	Not included	Not included	No confirmed detections worldwide since March 2020; NACI endorsed trivalent formulations from 2024–25 [19,20].

2.2 Where the strains come from: the GISRS pipeline

The WHO Global Influenza Surveillance and Response System (GISRS) has run continuously since 1952 [16]. The sequence for a Northern Hemisphere recommendation is:

- **Specimen collection (September–January).** National Influenza Centres in >120 countries (in Canada, the National Microbiology Laboratory and provincial laboratories such as Alberta Precision Laboratories) type and subtype respiratory specimens and forward representative viruses to the WHO Collaborating Centres (Atlanta, London, Melbourne, Tokyo, Beijing, Koltsovo, Memphis). Sequences are shared through GISAID EpiFlu and tracked in real time on Nextstrain [17,21].
- **Genetic characterization.** HA (and NA) genes are sequenced and placed on phylogenetic trees; clades and subclades are defined by signature amino-acid substitutions (for example subclade K's S144N/N158D/I160K cluster) [17].
- **Antigenic characterization.** Haemagglutination-inhibition (HI) assays, supplemented by virus-neutralization for H3N2 viruses that no longer agglutinate red cells, test circulating viruses against panels of post-infection ferret antisera raised to current and candidate vaccine viruses. Ferrets are used because they are immunologically naive, so each antiserum reflects a single virus [17,22]. Results are integrated by antigenic cartography (University of Cambridge), a multidimensional-scaling method that places viruses on a map where distance equals fold-drop in titre; a ≥4-fold (2-unit) drop in titre against the current vaccine virus is the conventional signal that an update should be considered [23].

- **Human serology.** For 2026–27, 15 serum panels from children, adults and adults ≥65 who had received standard-dose, high-dose, adjuvanted, cell-based or recombinant 2025–26 vaccines were tested against new viruses; significantly reduced post-vaccination titres to D.3.1 H1N1, subclade K H3N2 and C.3.1 B/Victoria viruses were the human-level evidence for all three changes [17,24].
- **Modelling.** Independent groups supply fitness forecasts that predict which clades will dominate 9–12 months out, using clade growth rates, HA epitope mutation load and antigenic advance [25–27]; the Global Influenza Vaccine Effectiveness (GIVE) collaboration provides interim vaccine-effectiveness estimates [17].
- **Recommendation and manufacture (February → August).** The consultation selects prototype viruses; WHO Collaborating Centres and manufacturers generate candidate vaccine viruses (high-growth egg reassortants, or cell isolates for cell/recombinant platforms); the four Essential Regulatory Laboratories (TGA, MHRA, FDA/CBER, NIID) calibrate HA potency reagents; Health Canada then authorizes each manufacturer's annual formulation [16,17].

2.3 Why egg-based and cell-based vaccines list different viruses

Influenza viruses grown in embryonated eggs acquire egg-adaptive HA mutations that improve growth but can alter antigenicity. For H3N2 this has been linked to reduced vaccine effectiveness in several seasons [28,29]. The 2026–27 report makes the mechanism explicit for influenza B: every available egg isolate of subclade C.3/C.3.1 lost the D197N glycosylation site, and ferret antisera raised to the egg-propagated B/Tokyo/EIS13-175/2025 recognized circulating C.3.1 viruses less well than antisera to the cell-propagated B/Pennsylvania/14/2025 [17]. WHO therefore names the best-matched prototype for each platform. Flucelvax (MDCK cell culture) avoids egg adaptation; Fluzone, Fluviral, Fluad and Fluzone High-Dose are egg-grown [18,19]. NACI makes no platform preference for adults under 65 and treats any age-appropriate product as acceptable [19].

Patient-education points

• "This year's vaccine was updated for all three strains because the viruses that circulated last winter, especially a new H3N2 variant called subclade K, had drifted away from last year's vaccine."

• "The strains are chosen in February by WHO from viruses collected around the world, tested in the lab against antibodies from vaccinated people and animals, and forecast with computer models; the vaccine then takes about six months to make, which is why it is a prediction rather than a perfect match."

• "There is no B/Yamagata component because that strain has not been seen anywhere since 2020."

3. Fluzone High-Dose (prior 65+ product) versus Fluad (adjuvanted; current 65+ product)

3.1 What changed in Alberta

Alberta funded Fluzone High-Dose for all adults ≥65 from 2021–22 through 2024–25 (and for long-term-care residents ≥65 in 2020–21). In 2025–26 the province moved adults ≥65 to Fluad (MF59-adjuvanted trivalent) and retained Fluzone High-Dose, now trivalent, for hematopoietic stem-cell, CAR-T and solid-organ transplant candidates and recipients ≥18 years (an off-label use based on Alberta transplant-program expert opinion). The 2026–27 program continues that arrangement [18,30]. Alberta used Fluad for seniors previously, in 2015–16 through 2017–18 [18]. The policy documents do not state the reason for the 2025 change; because NACI rates the two products as equivalent-tier options (section 3.4), product choice at program level is a procurement decision.

	Fluzone High-Dose Trivalent (Sanofi)	Fluad (Seqirus)
Platform	Egg-grown, split-virion, inactivated, trivalent	Egg-grown, surface-antigen (subunit) inactivated, trivalent
Antigen per 0.5 mL	60 µg HA per strain (180 µg total; 4× standard dose)	15 µg HA (with NA) per strain (standard antigen dose)
Adjuvant	None	MF59C.1: squalene oil-in-water emulsion stabilized by polysorbate 80 and sorbitan trioleate (≈9.75 mg squalene)
Health Canada indication	≥65 years (quadrivalent HD authorized 2020; trivalent HD in use from 2025–26)	≥65 years (in Canadian programs since 2011–12)
Alberta funding 2026–27	HSCT, CAR-T and SOT candidates/recipients ≥18 years (preferred product for this group); not the routine 65+ product	All adults ≥65 years
Dose / route	0.5 mL IM, 1 dose	0.5 mL IM, 1 dose
Appearance	Colourless opalescent suspension; shake well	Milky-white emulsion; gently shake
Interchangeability	Any inactivated influenza vaccine may be interchanged for persons ≥65 [18]	Same
Expected reactogenicity vs standard dose	More injection-site pain; grade ≥3 local events RR 5.0 (95% CI 0.9–28.7, imprecise) [31]	More injection-site pain; grade ≥3 local events RR 3.4 (1.3–8.7) [31]

3.2 Why older adults need an enhanced vaccine, and how each product addresses it

Immunosenescence (fewer naive T cells, weaker T follicular helper and germinal-centre function, blunted innate signalling) lowers both seroconversion and effectiveness of standard-dose vaccine: adults 17–59 mount roughly 2–4-fold higher antibody responses than adults ≥65, and pooled vaccine effectiveness against laboratory-confirmed influenza is about 49% in older adults versus 59% in younger adults [31].

- **High-dose** compensates by quadrupling antigen, which drives higher HI titres across strains; the pivotal FIM12 trial (n = 31,989) showed 24.2% (95% CI 9.7–36.5) relative efficacy against laboratory-confirmed influenza versus standard dose [32].
- **MF59** compensates by changing the quality of the innate response at the injection site: it rapidly recruits neutrophils, monocytes and dendritic cells, enhances antigen uptake and transport to the draining lymph node, and increases germinal-centre and T follicular helper activity, producing higher titres and broader cross-reactivity to drifted strains with no more antigen [33,34]. Adjuvant-driven breadth is the reason the human serology panels used by WHO include adjuvanted-vaccine recipients

[17].

3.3 Evidence against standard dose

- **High-dose:** FIM12 (above) [32]; the 2025 pragmatic registry trials DANFLU-2 (Denmark, n = 332,438) and GALFLU (Spain, n = 133,882) and their pre-specified pooled analysis FLUNITY-HD (Lancet 2025) found relative effectiveness against hospitalization for influenza or pneumonia of 8.8% (95% CI 1.7–15.5; DANFLU-2 alone 5.9%, not significant; GALFLU 23.7%), against laboratory-confirmed influenza hospitalization 31.9% (19.7–42.2), and against all-cause hospitalization 2.2% (0.3–4.1); all-cause mortality did not differ. Number needed to vaccinate with HD rather than SD to prevent one influenza/pneumonia hospitalization ≈ 1,839 [35,36]. Earlier observational Medicare data suggested a 31% reduction in influenza-associated deaths [31].
- **Adjuvanted:** the single RCT versus standard dose used a clinical ILI endpoint and showed no difference (relative efficacy −3%, 95% CI −19 to 11); the supporting evidence is observational, with pooled relative effectiveness against outpatient/ED influenza visits ranging from 0% to 36% depending on study design, and consistent reductions in influenza-coded hospitalizations across seven observational studies [31,37].

3.4 Head-to-head evidence (the question that matters for Alberta's switch)

- **Cluster-randomized crossover study, Kaiser Permanente Northern California, 2023–24 season** (JAMA Netw Open 2026): 429,595 adults ≥65 received adjuvanted (n = 212,875) or high-dose (n = 216,720) vaccine by facility allocation. Relative effectiveness of adjuvanted versus high-dose was 1.5% (95% CI −8.4 to 10.5) against PCR-confirmed influenza, 9.1% (−4.0 to 20.4) against influenza with hospitalization or ED visit, and 1.0% (−11.4 to 12.0) against community-acquired-pneumonia hospitalization; non-inferiority P <0.001 for all. The authors concluded the two products can be used interchangeably in this age group [38].
- **Randomized immunogenicity trial** (Immunity & Ageing 2023): 680 adults ≥65 randomized to trivalent adjuvanted or high-dose vaccine over two seasons; HD seroconversion for A(H3N2) was not statistically superior to the adjuvanted product [39]. A randomized trial in long-term-care residents found the adjuvanted vaccine non-inferior for anti-neuraminidase responses, with HI differences varying by season [40].
- **NACI/ACIP meta-analysis of four observational studies** (four seasons): influenza-associated hospitalization rVE for HD versus adjuvanted 4% (95% CI −1 to 10); outpatient/ED visits −6% (−23 to 8) — no meaningful difference [31,37]. The one RCT comparing all three enhanced products (HD, adjuvanted, recombinant) was far too small to discriminate (rVE point estimates with CIs spanning −213% to 96%) [31].
- **Safety head-to-head:** HD may cause fewer grade ≥3 systemic reactions than adjuvanted (RR 0.73, 95% CI 0.29–1.80) and fewer serious adverse events (RR 0.65, 0.32–1.30), but both estimates are imprecise; HD was associated with fewer SAEs than standard dose (RR 0.91, 0.85–0.97) [31]. No GBS signal distinguishes the products [31].

NACI's position (supplemental guidance, July 2024; reaffirmed in the 2026–27 statement): high-dose, adjuvanted or recombinant vaccine "should be offered, when available, over other influenza vaccines for adults 65 years of age and older," with no preference among the three; NACI's earlier (2018) preference for high-dose was framed against standard dose and has not been extended to a ranking among enhanced products, because head-to-head data show no difference [19,31]. The US ACIP has held the same position since 2022–23 [37].

3.5 Practical differences for the pharmacist

- Co-administration: influenza vaccines, including Fluad, may be given with any other vaccine. NACI notes that data on giving MF59-adjuvanted vaccine with AS01-adjuvanted vaccines (Shingrix, Arexvy) are limited, though Alberta policy permits same-day administration [18,31].

- Transplant recipients ≥18: Fluzone High-Dose remains Alberta's preferred product, including in pregnancy and lactation with informed consent about limited data; Fluad has no pregnancy data and is not indicated under 65 [18].
- Egg allergy is not a contraindication for either product; observe 30 minutes [18].
- Expect more arm soreness than with standard-dose vaccine with either product; systemic reactions are similar [31].

Patient-education script

"Both the high-dose shot you had before and this year's Fluad are 'enhanced' flu vaccines for people 65 and over. High-dose works by giving four times the amount of flu protein; Fluad gives the standard amount but adds an ingredient called an adjuvant that makes the immune system respond more strongly and to a wider range of strains. In a study of more than 400,000 seniors the two performed the same at preventing flu and flu hospitalizations, and Canada's advisory committee considers them equally good choices. Your arm may be a bit more sore for a day or two than with a regular flu shot, which is normal."

4. RSV vaccines for adults: what they are, why no booster yet, and what may change

4.1 Pharmacology

RSV enters cells through its fusion (F) glycoprotein, which exists on the virion in a metastable **prefusion** conformation and springs irreversibly into a stable **postfusion** shape during entry. Decades of failed subunit vaccines used postfusion-like F. The breakthrough was structural: McLellan and colleagues solved the prefusion F structure bound to a neutralizing antibody (Science 2013) and then engineered disulfide and cavity-filling mutations (DS-Cav1) that lock F in the prefusion state [41,42]. In preclinical studies stabilized prefusion F elicited neutralizing titres many-fold higher than postfusion F, and prefusion-specific antibodies (sites Ø and V) account for most of the neutralizing activity in human serum [42,43]. All three authorized adult vaccines are built on this principle [44].

	Arexvy (GSK)	Abrysvo (Pfizer)	mResvia (Moderna)
Antigen	RSVPreF3: 120 µg recombinant stabilized prefusion F (RSV-A derived)	RSVpreF: bivalent, 60 µg prefusion F from RSV-A + 60 µg from RSV-B	mRNA-1345: 50 µg mRNA encoding membrane-anchored stabilized prefusion F, in lipid nanoparticle
Adjuvant	AS01E (MPL + QS-21 in liposomes) — same adjuvant family as Shingrix	None	None (LNP is intrinsically immunostimulatory)
Canadian authorization	≥60 years; 50–59 years at increased risk	≥60 years; 18–59 years at increased risk; pregnancy 32–36 weeks (infant protection)	≥60 years; 18–59 years at increased risk
Pivotal efficacy (season 1, RSV-LRTD)	82.6% (96.95% CI 57.9–94.1); 94.1% against severe LRTD [45]	66.7% (≥2 symptoms); 85.7% (≥3 symptoms) [46]	83.7% (≥2 symptoms); 82.4% (≥3 symptoms) [47]
Durability	Cumulative 62.9% over 3 seasons (season 3 alone 48.0%) [48,49]	88.9% and 77.8% (≥3 symptoms) cumulative at end of seasons 1 and 2 [50]	≈50% (37.5–60.7) against ≥2-symptom LRTD through 24 months; faster waning than protein vaccines noted by ATAGI [51,52]
Safety signal	Rare GBS signal ≥65 years (both protein vaccines); benefit judged to outweigh [44]	As Arexvy	No GBS signal identified to date; less experience [44]

4.2 Current recommendations

NACI (April 2026) strongly recommends a single dose for all adults ≥75 and for adults 65–74 at increased risk, and for adults ≥18 who are nursing-home residents, lung- or stem-cell-transplant recipients, on home oxygen, or on dialysis; any of the three vaccines may be used. It states that one dose protects for at least three years and that "it is not yet known if the immune responses to these vaccines can be boosted with further doses," advising healthy adults who are not yet eligible to defer rather than vaccinate early [44,53]. **ACIP** likewise recommends a single dose (≥75, and 50–74 at increased risk) and states that previously vaccinated persons should not receive another dose; additional doses will be evaluated as data accrue [54,55]. **Alberta** funds one lifetime dose for community-dwelling adults ≥70, adults ≥60 in continuing-care or supportive-living settings, First Nations, Métis and Inuit adults ≥60, and, from 1 October 2026, adults 18–69 with lung or stem-cell transplant, chronic oxygen therapy or dialysis; the provincially supplied product has been Abrysvo (confirm current supply on the AIP biological page) [56,57].

4.3 Why there is no booster recommendation

- **A single dose is still working.** The only vaccine with three-season randomized data (AReSVi-006, n ≈ 25,000) showed cumulative efficacy of 62.9% (97.5% CI 46.7–74.8) against RSV-LRTD and 67.4% against severe LRTD, with efficacy declining from 82.6% in season 1 to 48.0% in season 3 [48,49]. RSVpreF retained 77.8% through two seasons [50].
- **Early revaccination added nothing.** In AReSVi-006, vaccine recipients were re-randomized before season 2 to a second dose or placebo. Two-season efficacy was 67.2% with one dose versus 67.1% with annual revaccination; three-season efficacy 62.9% versus 67.8%, with overlapping intervals and no clinically meaningful gain [48,49]. The antibody rise after a 12-month second dose was smaller than after the first dose, consistent with pre-existing antibody blunting ("epitope masking") the response [48,58].
- **No trial shows a clinical benefit of a second dose at any interval.** Immunogenicity studies exist — RSVPreF3 revaccination at 18 months after a two-dose primary series was immunogenic [59]; mRNA-1345 revaccination at 12 and at 24 months produced neutralizing titres non-inferior to the primary dose [60,61] — but titre non-inferiority does not establish incremental protection, and no correlate of protection has been validated [44].
- **Each dose carries a rare but real risk.** Post-marketing surveillance identified an elevated risk of Guillain-Barré syndrome in adults ≥65 after the protein-subunit vaccines (on the order of several excess cases per million doses in US surveillance). NACI judged the benefit of a first dose to outweigh it; a second dose must demonstrate its own benefit before that calculus is repeated [44,53].
- **Programmatic prudence.** Advisory committees set schedules on efficacy outcomes, not antibody curves; the first products were authorized only in 2023, so the longest controlled follow-up is three seasons.

4.4 Will this change?

Very probably, but toward a **spaced revaccination** rather than an annual schedule. The signals pointing that way:

- Documented waning: 82.6% → 48.0% by season 3 for RSVPreF3 [49]; roughly 50% by 18–24 months for mRNA-1345, with modelled efficacy near 30% at 36 months [51,52]. Protection in a fourth or fifth season is unknown.
- Longer-interval trials are under way: GSK's persistence study randomizes revaccination at 24, 36, 48 and 60 months (NCT04732871) [62]; Moderna has 12- and 24-month revaccination data and regulators (e.g., Australia's TGA) have made further revaccination data a condition of approval [52,60,61]; Pfizer is evaluating 12- and 24-month intervals [44].
- Mechanistic rationale for waiting: antibody levels fall over 2–3 years, and the mRNA-1345 investigators note that lower pre-existing titres at longer intervals should permit a full booster response [61]; the 12-month data from AReSVi-006 show the converse [48].
- Both NACI and ACIP have explicitly committed to revisiting revaccination as efficacy data at longer intervals emerge [44,54,55]; GSK has stated it will supply longer-term data to inform schedules [49].

A reasonable expectation is a recommendation for a second dose at an interval of several years, possibly prioritized by age (≥75–80) or frailty, once a randomized or large observational study shows that revaccination restores protection against LRTD or hospitalization. Until then, the correct counselling is that one dose is current standard of care and that a patient who has had one should not receive another, regardless of product.

Patient-education script

"The RSV vaccine teaches your immune system to recognize the protein the virus uses to get into your cells, in the exact shape it has before it attacks — that's why it works so much better than older attempts. One dose has been shown to cut serious RSV lung infections by about 60% over three winters. A second dose a year later did not add protection in the trials, so right now only one dose is recommended. Researchers are testing boosters at two to five years, and the advice will likely change once those results are in; we will let you know if you become eligible for another dose."

5. Patient counselling sheet: how long does flu and COVID-19 vaccine protection last?

Each item pairs a plain-language script (say it almost verbatim at the counter) with the pharmacist-level detail behind it for the patient who asks "why?". Added 2 November 2026.

5.1 Influenza vaccine

What to tell patients

"Your flu shot takes about two weeks to reach full strength, and then protection slowly fades over the winter, by roughly 7 to 10 percent each month. A shot in October still gives you solid protection through January and February, when flu usually peaks here. That is why we recommend one dose each fall, and why we don't suggest waiting until later to get it."

Why it fades (pharmacist detail)

- Antibody titres peak at 2–4 weeks and then decline. In the US Flu VE Network, vaccine effectiveness fell about 7% per month for A(H3N2) and B and 6–11% per month for A(H1N1)pdm09 [63]; Kaiser Permanente data showed a 16% rise in the odds of testing positive for every 28 days since vaccination [64].
- Waning is fastest for H3N2 and in adults ≥65 (immunosenescence). This is relevant in 2026–27 because subclade K H3N2 drove last season and all three vaccine strains were updated (section 2).
- NACI's 2026–27 review concluded that early-fall vaccination still covers most of the season; delaying to chase peak titres is not warranted, and coverage, not timing, is the main determinant of program impact [19].

For patients 65+ asking why they get a different shot

"Fluad is designed for people 65 and over. It includes an ingredient called an adjuvant that helps your immune system respond more strongly and more broadly, so you start from a higher level of protection and it holds up better through the season."

Detail: MF59 recruits innate immune cells at the injection site, improves antigen transport to the draining node and boosts germinal-centre activity, giving higher and broader titres with standard antigen content [33,34]. Head-to-head with high-dose, effectiveness is equivalent (rVE 1.5%, 95% CI −8.4 to 10.5; n ≈ 430,000) [38].

5.2 COVID-19 vaccine

What to tell patients

"The COVID vaccine is best at preventing serious illness, hospital stays and death. That protection is strongest in the first two months and gradually drops over about four to six months. Protection against catching a mild infection at all is shorter, a few months at most. That is why the fall dose is timed just before the winter season, and why people at higher risk, such as those 80 and over, are offered a second dose a few months later."

Why it fades (pharmacist detail)

- Against hospitalization or death, the 2024–25 vaccines gave about 57% protection at 4 weeks, falling to 34% at 20 weeks, in a Nebraska population cohort [65]; in the US VISION network, effectiveness against hospitalization was 51% (95% CI 43–58) at 7–59 days and no longer measurable (0%, −15 to 13) at 180–299 days in immunocompetent adults [66].
- Against infection, protection was about 45% at 1 month and 17% by 5 months in the same cohort [65]. Neutralizing antibody wanes faster than T-cell memory, so protection against mild infection goes

first and protection against severe disease lasts longer.

- Antigenic drift adds to the decline; the updated formulation each fall partly resets the match. NACI states that protection wanes after about 4–6 months [67].

Second dose: who and when (NACI guidance for fall 2026) [67]

- Strongly recommended as a second dose in the year: adults ≥80; residents of long-term care and other congregate senior living; moderately to severely immunocompromised people ≥6 months; adults 65–79 with conditions that raise the risk of severe COVID-19. May be offered to all other adults 65–79.
- Minimum interval 3 months from the previous dose. Alberta's funded fall program covers adults ≥65, LTC/congregate residents, people with underlying or immunocompromising conditions, pregnancy, First Nations/Inuit/Métis individuals and health-care workers, with at least 3 months since the last dose [30].

5.3 One-line comparison for the counter

	Influenza vaccine	COVID-19 vaccine
Full strength by	About 2 weeks	About 2 weeks
How protection changes	Fades steadily, roughly 7–10% per month	Strong for about 2 months, then declines
Still useful at	4–6 months (through the season)	4–6 months for severe disease; protection against infection is shorter
How often	Once each fall	Once each fall; second dose ≥3 months later for higher-risk groups
Best timing	October–November	Just before winter; second dose in spring for those eligible

6. Patient counselling sheet: "What strains are in the flu vaccine this year?"

Layered answer: the 20-second version for most patients, the names if they ask, the common follow-ups, and the pharmacist-level background. Added 2 November 2026.

6.1 The 20-second answer (most patients)

What to tell patients

"This year's vaccine covers three flu strains: two influenza A strains, an H1N1 and an H3N2, and one influenza B strain. All three were updated this year because the viruses drifted quite a bit last winter, especially the H3N2 strain that caused the big season. So even if you had a flu shot last year, this one is different and worth getting."

The three things that matter to a patient: **it covers A and B**, **all three strains are new**, and **last year's shot does not cover this year's viruses well**.

6.2 If they want the actual names

What to tell patients

"A/Missouri/11/2025 (H1N1), A/Darwin/1454/2025 (H3N2), and B/Tokyo/EIS13-175/2025, a B/Victoria strain. If you're getting Flucelvax, the cell-based one, two of the names are slightly different, A/Darwin/1415/2025 and B/Pennsylvania/14/2025, but they're the same strains grown a different way."

Decoding the name for them: type / city where that virus was first isolated / lab number / year. "A/Darwin/1454/2025" is influenza A, sample 1454, isolated in Darwin, Australia, in 2025. "-like" on the official list means the vaccine uses a virus antigenically matching that one.

6.3 Common follow-up questions

"Last year had four strains. Why only three?"

"The fourth one, a B strain called Yamagata, hasn't been detected anywhere in the world since March 2020. Keeping it in wasn't protecting anyone, so it was removed."

"Does the vaccine match what's going around?"

"The strains were chosen in February from viruses collected worldwide, and the H3N2 component specifically targets the variant called subclade K that caused last year's early, heavy season. Flu keeps changing, so no vaccine is a perfect match, but even partial matches reduce how sick you get and keep people out of hospital."

"Can I get the flu from the shot?"

"No. Every injectable flu vaccine in Alberta is inactivated; there's no live virus. A sore arm and feeling run down for a day are your immune system responding, not an infection."

"I'm allergic to eggs."

"You can still get any of them, including the egg-based ones; egg allergy isn't a contraindication. Flucelvax is grown in cells rather than eggs if you'd prefer it."

"Which one am I getting?"

- Under 65: Fluzone, Fluviral or Flucelvax (standard dose, 15 µg HA per strain; any is fine).
- 65 and over: Fluad (MF59-adjuvanted; standard antigen, stronger and broader response).
- Transplant candidates/recipients ≥18 (HSCT, SOT, CAR-T): Fluzone High-Dose (60 µg HA per strain).

6.4 Pharmacist-level background

What the strain changes actually mean

Component	2025–26 (egg)	2026–27 (egg)	Clade change	Why it changed
A(H1N1)pdm09	A/Victoria/4897/2022	A/Missouri/11/2025	5a.2a.1 → subclade D.3.1	D.3.1/D.3.1.1 now predominate; post-vaccination human titres were significantly reduced against them [17].
A(H3N2)	A/Croatia/10136RV/2023	A/Darwin/1454/2025	J.2 → subclade K (formerly J.2.4.1)	Emerged Aug 2025; 7 HA substitutions incl. S144N (new glycosylation site), N158D, I160K. Croatia-strain ferret antisera barely recognized K (HI <40 vs 640–1280 homologous) [17].
B/Victoria	B/Austria/1359417/2021	B/Tokyo/EIS13-175/2025	3a.2 → subclade C.3.1	D197N (new glycosylation site); rising in North America and East Asia; poorly recognized by B/Austria antisera [17].

Why egg and cell products list different viruses. Egg-grown viruses acquire adaptive HA mutations that can change antigenicity, documented for H3N2 in 2012–13 and 2016–17 [28,29]. This year the WHO report shows it for influenza B: every egg isolate of subclade C.3.1 lost the D197N glycosylation site, so a cell-grown prototype (B/Pennsylvania/14/2025) was named for Flucelvax and recombinant products [17]. Clinical differences are small; NACI has no platform preference under 65 [19].

What "match" is worth, with real numbers. Last season is the teaching case: subclade K caused a drifted-H3N2 season against a vaccine that did not contain it, and interim test-negative estimates still showed meaningful protection. In Canada (AB, BC, ON, QC outpatients, 26 Oct 2025–10 Jan 2026; 87% of sequenced H3N2 were subclade K), VE against H3N2 was 40% overall (95% CI 28–49) and 37% (20–50) against sequenced subclade K viruses; by age, 36% (1–17 y), 48% (18–64 y) and 25% (−7 to 48) in ≥65 [68]. Beijing reported 23.5% (11.7–33.7) against laboratory-confirmed influenza in a K-dominated season [69]. That is what "mismatch" looks like: reduced, not zero. The K strain is in this year's vaccine, so better H3N2 protection is expected, but no 2026–27 effectiveness data exist yet; Canadian interim estimates usually appear in late January.

How the selection was made (short version). GISRS laboratories in more than 120 countries sent viruses to WHO Collaborating Centres from September to January; HA genes were sequenced and placed in clades; HI and neutralization assays with post-infection ferret antisera measured antigenic distance; 15 human serum panels from people vaccinated with the 2025–26 products (including Fluad and high-dose recipients) showed significantly reduced titres to the new H1N1, K and C.3.1 viruses; fitness models forecast which clades would dominate; WHO issued the recommendation on 27 February 2026. Manufacturers then needed about six months, which is why the vaccine is always a February prediction about November [16,17]; section 2.2 gives the full pipeline. **Where to look things up:** exact reassortant viruses per product ("IVR-"/"NYMC X-" strains) are in the Health Canada product monograph; "-like" prototypes on the AIP biological pages [18]; current circulation on FluWatch and the Alberta respiratory virus dashboard.

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Prepared as an educational reference for pharmacist practice; it does not replace the Alberta Immunization Policy, product monographs or clinical judgment. Verify provincial eligibility and product supply against the current AIP biological pages before administration.